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1. Metrics Glossary

- **Decision Margin (DM):** An internal confidence score reported by the algorithm.
 - Higher DM → the detector/tracker is more certain about its estimate.
 - Lower DM → the estimate may be unstable or close to losing track.
- **Frame Jitter (ms):** Variability in frame-to-frame processing time.
 - Low jitter → smoother frame delivery and stable performance.
 - High jitter → irregular updates, leading to uneven motion tracking.
- **Orientation Jitter (°):** Angular noise in orientation estimates.
 - Low jitter → head/camera rotation looks smooth.
 - High jitter → even small movements appear shaky.
- **Translation Jitter (mm):** Positional noise between frames.
 - Low jitter → virtual objects stay fixed in space.
 - High jitter → objects appear to “swim” or vibrate.
- **Drift (Position, mm):** Offset between the start and end position during a test.
 - Low drift → stable over time.
 - High drift → gradual creeping away from the correct pose.
- **Drift (Angle, °):** Rotational offset between the start and end of a test.
 - Low drift → orientation remains consistent.
 - High drift → alignment gradually drifts.
- **Beta X / Beta Y / Beta Z (°):** Supported head rotation ranges on each axis during the test.
 - Higher ranges → system successfully tracks wider head rotations.
 - Lower ranges → system struggles when the head rotates far.
- **Relock Time (Median, ms):** The median time required for the system to regain stable tracking after a temporary loss.
 - Lower median → quick recovery after tracking loss.
 - Higher median → slower re-acquisition of the target.
- **Δt (95th Percentile, ms):** The maximum frame interval below which 95% of frame updates occur.
 - Lower value → more consistent frame delivery with fewer long gaps.
 - Higher value → occasional long frame delays, which can impact smoothness.

2. Tests

A dedicated testing environment was created to evaluate the performance of our tracking systems. The environment operates by collecting pose data through the CSV output generated by the marker-tracking code (FaceMesh and AprilTag based pipelines). These CSV logs contain positional, rotational, and confidence-related metrics for each frame, which are later analyzed to extract performance indicators such as jitter, drift, and Decision Margin. The tests were conducted with 2 person for every motion scenario for better data collection.

The tests were designed to represent three distinct motion scenarios:

1. **Stationary Test:** The subject's head remains still to measure inherent stability, noise, and long-term drift without movement influence.
2. **Stable Motion Test:** The subject performs slow, controlled head movements to evaluate smoothness, continuity, and robustness of tracking during normal use.
3. **Rapid Motion Test:** The subject performs quick and abrupt head movements to stress-test the trackers under rapid changes, measuring recovery time, relock performance, and motion robustness.

This structured testing framework allows us to directly compare algorithm performance under realistic conditions, while still isolating different aspects of stability and responsiveness.

2.1 Stationary Test

Test Setup: The stationary test was performed by keeping the subject's head still in front of the camera. The goal of this test is to evaluate stability-related metrics such as jitter, drift, and confidence (Decision Margin), without the influence of movement. This allows us to measure the inherent noise, stability, and long-term consistency of each tracking algorithm.

Metrics Observed:

Orientation Jitter (°): Angular noise when the head is not moving. Lower values = smoother orientation tracking.

Translation Jitter (mm): Positional noise when stationary. Lower values = pose does not "vibrate."

Drift (Position, mm): Offset between start and end despite no movement. Lower = better long-term stability.

Drift (Angle, °): Rotational offset between start and end. Lower = orientation remains consistent.

Decision Margin (DM): Internal confidence score. Higher = algorithm is more certain about its output.

	Facemesh Data	AprilTag Data	Conclusion
Orientation jitter	0.331	0.556	%40.6 FaceMesh
Translation jitter	4.769	0.741	%84 AprilTag
Drift pose (mm)	23.595	2.945	%87.5 AprilTag
Drift angle	5.061	0.438	%91.3 April Tag
Decision margin (median)	126	63	%101 FaceMesh

Table-1 Stationary Datas of Person1

	Facemesh Data	AprilTag Data	Conclusion
Orientation jitter	0.356	1.742	%79.5 FaceMesh
Translation jitter	3.738	0.671	%81 AprilTag
Drift pose (mm)	38.098	3.647	%90.4 AprilTag
Drift angle	5.206	3.323	%36.2 April Tag
Decision margin (median)	119	62	%92 FaceMesh

Table-2 Stationary Datas of Person2

2.1.1 Summary Considering Filtering and Interpolation

Both FaceMesh and AprilTag tracking systems were evaluated under stationary conditions. The raw results show that FaceMesh achieves lower orientation jitter and higher Decision Margin (confidence), while AprilTag produces much lower translation jitter and significantly less drift over time.

However, it is important to note that **positional and angular jitter can be strongly prevented by applying post-processing methods such as a Kalman filter or interpolation before sending them via UDP**. These techniques effectively smooth out frame-to-frame noise and make orientation and translation appear more stable to the end user. And also the results may be affected because of the personal errors made while testing.

As a result:

- **Jitter differences (orientation/translation)** between FaceMesh and AprilTag become less critical after filtering, since both systems can be smoothed to a comparable stability level.
- **Drift** (long-term positional or angular offset) cannot be fully corrected by filtering or interpolation, because it represents a systematic shift rather than random noise. Here, AprilTag consistently shows an advantage.
- **Decision Margin** remains a useful metric even after filtering, since it reflects the underlying confidence of each algorithm's raw detections. FaceMesh maintains higher DM, which may translate into more robust performance under challenging conditions (occlusion, partial views), even if drift remains an issue.

2.1.2 Final Results

AprilTag achieves lower translation jitter and drift, showing better long-term positional stability. However, since these errors are on the order of millimeters, they are often imperceptible in practice and can be further reduced with Kalman filtering or interpolation. FaceMesh provides lower orientation jitter and much higher Decision Margin, meaning smoother rotational tracking and stronger confidence. Because angular noise is far more noticeable to the user than sub-mm drift, these strengths have greater impact in real AR/VR usage.

Conclusion: AprilTag is preferable when precise, drift-free anchoring is critical, while FaceMesh is generally the better choice for interactive scenarios where smoothness, rotational stability, and robustness are more important.

2.2 Stable Motion Test

Test Setup: In the stable motion test, the subject performed **smooth and controlled head movements** (left–right, up–down, and slight tilts). This test measures how well each algorithm maintains stability and accuracy when tracking normal user motions, without the stress of abrupt changes. The focus is on **frame stability, supported rotation ranges, and decision confidence**.

Metrics Observed:

- **Frame Jitter (ms):** Measures stability of processing time between frames. Lower = more consistent tracking.
- **Decision Margin (DM):** Algorithm’s confidence. Higher = more reliable detections during motion.
- **Beta X / Y / Z (°):** Effective supported head rotation ranges on each axis. Larger = better coverage of natural movements.

	Facemesh Data	AprilTag Data	Conclusion
Frame jitter	120.553	87.757	%27.6 AprilTag
Decision Margin	136	63	%117.6 FaceMesh
Beta X	104	74	%40 FaceMesh
Beta Y	11.5	89	%30 FaceMesh
Beta Z	360	360	-

Table-3 Stable Motion Datas of Person1

	Facemesh Data	AprilTag Data	Conclusion
Frame jitter	65.548	108.948	%39.8 FaceMesh
Decision Margin	118	57	%104.6 FaceMesh
Beta X	114	65.5	%74 FaceMesh
Beta Y	122	118	%3 FaceMesh
Beta Z	360	360	-

Table-4 Stable Motion Datas of Person2

2.2.1 Summary Considering Filtering and Interpolation

- **Jitter (frame/translation/orientation):** Post-processing filters can smooth frame-to-frame irregularities, reducing practical differences. This makes Frame Jitter less critical after smoothing.
- **Decision Margin:** Cannot be filtered away. FaceMesh’s consistently higher DM indicates stronger robustness under normal motion, which is valuable when the head is partially occluded or moving in low-light conditions.
- **Rotation Coverage (Beta ranges):** Determines how far the system can follow head motion before losing track. Here, FaceMesh demonstrates stronger performance, especially on the X axis.

2.2.2 Final Results

- **AprilTag strengths:** In Person 1, AprilTag showed lower frame jitter and stronger vertical tracking.
- **FaceMesh strengths:** In both persons, FaceMesh delivered much higher confidence, smoother jitter in one case, and significantly wider rotation support on horizontal and diagonal axes.

Conclusion: In slow, controlled head motions, **FaceMesh provides more reliable and smoother tracking with higher robustness**, while AprilTag occasionally achieves better frame-time stability. Overall, FaceMesh is the more suitable option for normal user movement scenarios.

2.3 Rapid Motion Test

Test Setup: In the rapid motion test, the subject performed **quick and abrupt head movements** (fast turns, nods, and shakes). This test stresses the tracking systems to their limits and evaluates their ability to maintain stability, quickly recover after loss, and handle sudden changes. The focus is on **relock performance, responsiveness, and robustness under high dynamics**.

Metrics Observed:

- **Relock Median (ms):** Measures how quickly the tracker recovers after losing the target. Lower = faster recovery.
- **Δt_{95p} (ms):** The 95th percentile of frame-to-frame time differences. Lower = fewer outlier delays, better real-time performance.
- **Decision Margin (DM):** Internal confidence score. Higher = more reliable detections even under fast motion.

	Facemesh Data	AprilTag Data	Conclusion
Relock Median	719.305	462.150	%35.8 AprilTag
Δt_{95p}	47.827	50.730	%5.7 FaceMesh
Decision Margin	149	59.647	%149.7 FaceMesh

Table-5 Rapid Motion Datas of Person1

	Facemesh Data	AprilTag Data	Conclusion
Relock Median	298.562	477.452	%7.5 FaceMesh
Δt_{95p}	47.674	412.693	%88.4 FaceMesh
Decision Margin	126.397	60.003	%110.7 FaceMesh

Table-5 Rapid Motion Datas of Person2

2.3.1 Summary Considering Filtering and Interpolation

- **Relock performance:** Filtering cannot improve relock directly, since this depends on detection robustness after loss. Here, AprilTag performed better in one case, while FaceMesh was faster in the second test, showing more variable but competitive results.
- **Timing stability (Δt_{95p}):** FaceMesh consistently provided smoother high-percentile frame times, which means fewer extreme delays.
- **Decision Margin:** Again, FaceMesh showed much higher confidence, which is especially important under fast motion where tracking is most likely to fail.

2.3.2 Final Results – Rapid Motion

- **AprilTag strengths:** Faster relock in Person 1, showing robustness in one of the tests.
- **FaceMesh strengths:** Much higher Decision Margin, smoother timing across both tests, and faster relock in Person 2.

Conclusion: Under rapid head movements, **FaceMesh is generally more reliable**, delivering higher confidence and smoother responsiveness, though AprilTag can occasionally recover faster. For real-world dynamic use cases, FaceMesh provides stronger overall robustness.

3. General Conclusion

Across all three test scenarios, both FaceMesh and AprilTag demonstrate useful strengths, but their practical implications differ:

- **Stationary Test:**
 - *AprilTag* showed lower translation jitter and drift, giving an edge in raw positional stability.
 - *FaceMesh*, however, achieved smoother orientation (lower angular jitter) and much higher Decision Margin, meaning it was more confident and visually stable. Since positional errors were in the millimeter range, these advantages are more meaningful for real user experience.
- **Stable Motion Test:**
 - *FaceMesh* consistently provided higher confidence and broader head-rotation coverage, especially on horizontal and diagonal axes.
 - *AprilTag* occasionally delivered lower frame jitter, but differences became negligible after filtering.
 - For normal, controlled movements, FaceMesh produced smoother, more reliable results overall.
- **Rapid Motion Test:**
 - *AprilTag* recovered faster in one test, showing resilience in a specific case.
 - *FaceMesh* provided higher Decision Margin in both tests, smoother timing, and faster relock in one case. This indicates that under fast, real-world head motions, FaceMesh is generally more robust.

Considering Filtering and Interpolation:

- Jitter-related noise can be strongly reduced by Kalman filtering and interpolation for both algorithms, reducing the raw differences.
- Drift remains a systematic error where AprilTag has an advantage.
- However, Decision Margin cannot be filtered away, and FaceMesh consistently provides higher confidence. This makes FaceMesh more robust under occlusions, partial detections, and dynamic scenarios.

Practical Factors:

- *AprilTag* requires an additional physical marker or equipment, which limits flexibility and can affect user comfort.
- *FaceMesh* works marker-free in any environment with just the camera feed, making it more practical and universal.
- In terms of effective reach and tracking coverage, both systems performed at similar levels, so the main differentiator is usability and robustness.

3.1 Final Recommendation

While AprilTag is a solid choice for setups where absolute positional stability and minimal drift are essential, its requirement for extra markers adds complexity. FaceMesh, being markerless, more confident, and smoother in both normal and fast-motion conditions, is overall the more practical and robust solution for AR/VR usage.